One-parameter maps

Vector-valued function: $\mathbf{r} : I \to \mathbb{R}^m$, $\mathbf{r}(t) = (r_1(t), \dots, r_m(t))$, where I is an interval in $\mathbb{R}$. The scalar t is the **parameter**; the scalar-valued r_i are its **component functions**. Its image traces a curve in $\mathbb{R}^m$.

Parametric line: $\mathbf{r}(t) = \mathbf{x}_0 + t\mathbf{d}$, where $\mathbf{x}_0 \in \mathbb{R}^m$, $\mathbf{d} \neq \mathbf{0}$, and $t \in \mathbb{R}$.

Limits and continuity

Limit: for $\mathbf{v} \in \mathbb{R}^m$,

$$\lim_{t \to a} \mathbf{r}(t) = \mathbf{v} \quad \iff \quad \lim_{t \to a} \|\mathbf{r}(t) - \mathbf{v}\| = 0.$$

Componentwise limits: if each scalar limit exists, then

$$\lim_{t \to a} \mathbf{r}(t) = \begin{bmatrix} \lim_{t \to a} r_1(t) \\ \vdots \\ \lim_{t \to a} r_m(t) \end{bmatrix}.$$

Continuity at a: $\mathbf{r}(a)$ is defined and $\lim_{t \to a} \mathbf{r}(t) = \mathbf{r}(a)$.

Theorem: $\mathbf{r}$ is continuous at a if and only if every component r_i is continuous at a.

Derivatives and tangent vectors

Derivative:

$$\frac{d\mathbf{r}}{dt}(t) = \mathbf{r}'(t) = \lim_{h \to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h},$$

provided the limit exists. Then $\mathbf{r}$ is **differentiable at** t. It is differentiable on an open interval if the derivative exists at every point of that interval.

Componentwise differentiation: $\mathbf{r}$ is differentiable at t if and only if every r_i is differentiable there; in that case,

$$\mathbf{r}'(t) = \begin{bmatrix} r_1'(t) \\ \vdots \\ r_m'(t) \end{bmatrix}.$$

Tangent vector: $\mathbf{r}'(t)$ at $\mathbf{r}(t)$; it gives a tangent direction when nonzero. The derivative is itself vector-valued.

Second derivative: $\mathbf{r}'' = (\mathbf{r}')'$, wherever it exists; differentiate each component again.

Differentiation rules

Let $\mathbf{u}, \mathbf{v} : I \to \mathbb{R}^m$ and $a : I \to \mathbb{R}$ be differentiable, and let $c \in \mathbb{R}$ be constant. All terms below are evaluated at the same parameter t.

$$(\mathbf{u} + \mathbf{v})' = \mathbf{u}' + \mathbf{v}', \qquad (c\mathbf{u})' = c\mathbf{u}',$$
$$(a\mathbf{u})' = a'\mathbf{u} + a\mathbf{u}', \quad (\mathbf{u} \cdot \mathbf{v})' = \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}'.$$